ON THE DISPERSION ANALYSIS OF THE META-IMPACTOR

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Abstract

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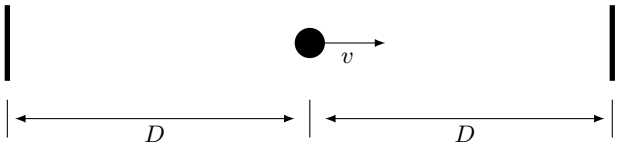


Figure 1: Particle travelling between rigid barriers at a constant velocity v.

1 Mathematical background

1.1 The Dirac comb and Fourier series

The periodic impulse train is defined as

$$w(t) = \sum_{\ell=-\infty}^{\infty} \delta(t - \ell T) \tag{1}$$

where $\delta(t)$ is the Dirac's delta function and T the period. The (complex) Fourier series of $w(t)$ is

$$w(t) = \frac{1}{T} \sum_{\ell=-\infty}^{\infty} e^{j2\pi \frac{\ell}{T} t} \tag{2}$$

For the time-scaled and time-shifted Dirac comb

$$w_s(t) = \sum_{\ell=-\infty}^{\infty} \delta(at \pm t_0 - \ell T) \tag{3}$$

Eq. 2 becomes (see App. A.1)

$$w_s(t) = \frac{1}{T} \sum_{\ell=-\infty}^{\infty} e^{j2\pi \frac{\ell}{T}(at \pm t_0)} \tag{4}$$

It is noted that the relationship between the coefficients of the complex Fourier series

$$x(t) = \sum_{\ell=-\infty}^{\infty} C_\ell e^{2\pi \frac{\ell}{T} t} \tag{5}$$

to their counterparts of the ordinary Fourier series expansion

$$x(t) = \frac{A_0}{2} + \sum_{\ell=1}^{\infty} A_\ell \cos\left(2\pi \frac{\ell}{T} t\right) + B_\ell \sin\left(2\pi \frac{\ell}{T} t\right) \tag{6}$$

is

$$C_\ell = \begin{cases} \frac{1}{2} A_0 & , \ell = 0 \\ \frac{1}{2}(A_\ell - jB_\ell) & , \ell = 1, 2, 3, \ldots \end{cases} \tag{7}$$

from which $A_\ell = 2\mathrm{Re}\{C_\ell\}$ and $B_\ell = -2\mathrm{Im}\{C_\ell\}$.

1.2 The non-smooth temporal transformation (NSTT)

1.2.1 The underlying physics

The motion of a rigid particle travelling between two rigid barriers of clearance $2D = 2$ m with *zero* initial displacement, *unit* initial velocity and *perfectly elastic collision*, is described by (see Fig. 1)

$$\tau(t) = \begin{cases} t & , -1 \leq t \leq 1 \\ 2 - t & , \quad 1 \leq t \leq 3 \end{cases} \tag{8}$$

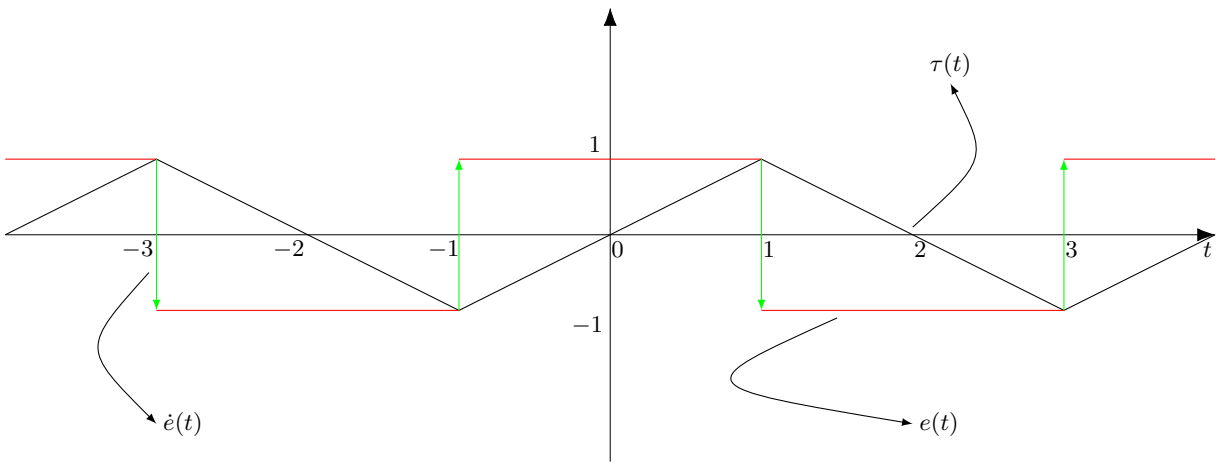


Figure 2: The functions $\tau(t)$ (black), $e(t) = \dot{\tau}(t)$ (red) and $\dot{e}(t)$ (green). The arrows in $\dot{e}(t)$ denote the sign of the associated impulse.

and illustrated in Fig. 2. Clearly, the function pertains to the standard triangular wave and it is periodic with period $T = 4$ s[1], i.e., $\tau(t+4) = \tau(t)$. The (distributional) derivative of $\tau(t)$ is

$$e(t) \equiv \dot{\tau}(t) = \begin{cases} 1 & , -1 \leq t \leq 1 \\ -1 & , \quad 1 \leq t \leq 3 \end{cases} \tag{9}$$

while it holds that $e^2(t) = 1$. Accordingly, from Fig. 2, the (distributional) derivative of $e(t)$ can be formed as

$$\dot{e}(t) = \cdots - 2\delta(t+3) + 2\delta(t+1) - 2\delta(t-1) + 2\delta(t-3) - \ldots$$

Expanding the latter on the left and right, to cover some additional terms not shown on the graph reads

$$\begin{aligned} \dot{e}(t) = \ldots - 2\delta(t+7) + 2\delta(t+5) - 2\delta(t+3) + 2\delta(t+1) \\ - 2\delta(t-1) + 2\delta(t-3) - 2\delta(t-5) + 2\delta(t-7)\ldots \end{aligned}$$

Factorizing with respect to the sign and accordingly re-arrange, in order to reveal the symmetries of the temporal shifts, leads to

$$\begin{aligned} \dot{e}(t) = \ldots 2\big\{\delta(t-7) + \delta(t-3) + \delta(t+1) + \delta(t+5)\big\} \\ -2\big\{\delta(t+7) - \delta(t+3) - \delta(t-1) - \delta(t-5)\big\}\ldots \end{aligned}$$

allowing thus to write $\dot{e}(t)$ in closed form as

$$\dot{e}(t) = 2\sum_{\ell=-\infty}^{\infty} \delta(t+1-4\ell) - \delta(t-1-4\ell) \tag{10}$$

In terms of trigonometric functions, the states of the impact oscillator (i.e., Eqs. 8–9) admit the following representation

$$\tau(t) = \frac{2}{\pi}\arcsin\left(\sin\left(\frac{\pi}{2}t\right)\right) \tag{11a}$$

$$e(t) = \operatorname{sgn}\left(\cos\left(\frac{\pi}{2}t\right)\right) \tag{11b}$$

where sgn is the sign function.

[1]There is no special reason of selecting [-1,3] for the fundamental period, other than the fact that $\tau(t)$ is represented by two branches.

The general case with $D \neq 1$ and $V \neq 1$ can still be expressed in terms of the above functions. The motion of the particle is now described by

$$x(t) = \begin{cases} Vt & , -\frac{D}{V} \leq t \leq \frac{D}{V} \\ 2D - Vt & , \quad \frac{D}{V} \leq t \leq 3\frac{D}{V} \end{cases} \tag{12}$$

and it is periodic, $x(t) = x(t + T_x)$, with $T_x = 4D/V = 4a$ (a is the quarter of the period). Equation 12 can be written as

$$x(t/a) = \begin{cases} D\frac{t}{a} & , -1 \leq \frac{t}{a} \leq 1 \\ D\left(2 - \frac{t}{a}\right) & , \quad 1 \leq \frac{t}{a} \leq 3 \end{cases} \tag{13}$$

Therefore

$$x(t) = D\tau(t/a) \tag{14}$$

with $\tau(t/a) = \tau(t/a + 4)$ and $T_\tau = 4a$. The derivatives of $x(t)$ read

$$\dot{x}(t) = D\dot{\tau}(t/a) \equiv \frac{D}{a}e(t/a) = \begin{cases} V & , -1 \leq t/a \leq 1 \\ -V & , \quad 1 \leq t/a \leq 3 \end{cases} \tag{15}$$

$$\ddot{x}(t) = D\ddot{\tau}(t/a) \equiv \frac{D}{a}\dot{e}(t/a) = 2V \sum_{i=-\infty}^{\infty} \delta(t/a + 1 - 4i) - \delta(t/a - 1 - 4i) \tag{16}$$

with $e^2(t/a) = V^2$, whereas the Eqs. 10–11 take the form

$$x(t) = \frac{2D}{\pi} \arcsin\left(\sin\left(\frac{\pi}{2}\frac{t}{a}\right)\right) = \frac{2D}{\pi} \arcsin\left(\sin\left(2\pi\frac{1}{4a}t\right)\right) = \frac{2D}{\pi} \arcsin\big(\sin(\omega t)\big) \tag{17a}$$

$$\dot{x}(t) = V\mathrm{sgn}\left(\cos\left(\frac{\pi}{2}\frac{t}{a}\right)\right) = V\mathrm{sgn}\left(\cos\left(2\pi\frac{1}{4a}t\right)\right) = V\mathrm{sgn}\big(\cos(\omega t)\big) \tag{17b}$$

$$\ddot{x}(t) = 2D \sum_{\ell=-\infty}^{\infty} \delta(t/a + 1 - 4\ell) - \delta(t/a - 1 - 4\ell) \tag{17c}$$

and depicted in Fig. 3(a) for $D = 2$ m and $V = 5$ m/s.

The set of Eqs. 17 can be further adjusted, to cover the most general case of $x(0) = x_0 \neq 0$. This may be succeeded by adding a phase argument in the inner sinusoidal and dirac functions as

$$x(t) = \frac{2D}{\pi} \arcsin\big(\sin(t/a + \phi)\big) \equiv D\tau(t/a + \phi) \tag{18a}$$

$$\dot{x}(t) = V\mathrm{sgn}\big(\cos(t/a + \phi)\big) \equiv Ve(t/a + \phi) \tag{18b}$$

$$\ddot{x}(t) = 2D \sum_{\ell=-\infty}^{\infty} \delta(t/a + 1 - 4\ell + \phi) - \delta(t/a - 1 - 4\ell + \phi) \tag{18c}$$

The exact value of ϕ is easily determined from Eq. 18a after setting $x(0) = x_0$:

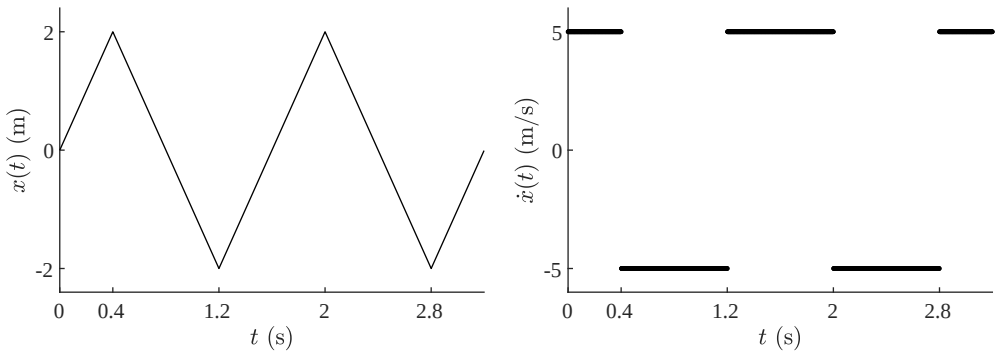
$$\phi = \frac{\pi}{2D}x_0 \tag{19}$$

Figure 3(b) illustrates the states of the oscillator for $x_0 = 1$ m.

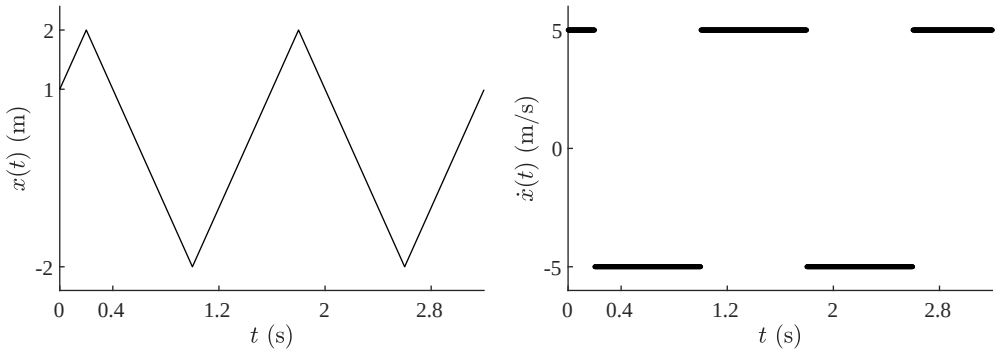
1.2.2 Some fundamental results

It is proven that

any periodic process $x(t)$ of period normalized to $T = 4$ can be expressed through the dynamic state of the



(a) $x_0 = 0$



(b) $x_0 = 1$ m

Figure 3: States of the impact oscillator for $D = 2$ m and $v = 5$ m/s ($a = 0.4$ s).

impact oscillator, $\{\tau(t), \dot{\tau}(t)\}$, in the form of a "hyperbolic complex number"

$$x(t) = X(\tau) + Y(\tau)e \tag{20}$$

where

$$X(\tau) = \frac{1}{2}\big[x(\tau) + x(2-\tau)\big] \tag{21a}$$

$$Y(\tau) = \frac{1}{2}\big[x(\tau) - x(2-\tau)\big] \tag{21b}$$

with $\tau = \tau(t)$, $e = e(t) = \dot{\tau}(t)$ and $e^2 = 1$.

<u>Example</u>: consider the functions

$$x_1(t) = A_1 \sin(2\pi f t)$$
$$x_2(t) = A_2 \cos(2\pi f t)$$

First we normalize the period to $T = 4$, by introducing a new variable θ as

$$2\pi f t = 2\pi \frac{1}{4}\theta \longrightarrow t = \frac{1}{4f}\theta \longrightarrow \theta = 4ft$$

so that

$$x_1(\theta) = A_1 \sin\left(\frac{\pi}{2}\theta\right)$$
$$x_2(\theta) = A_2 \cos\left(\frac{\pi}{2}\theta\right)$$

Then, applying Eqs. 20–21 to the new argument θ implies

$$X_1(\tau) = \frac{1}{2}\left[A_1 \sin\left(\frac{\pi}{2}\tau\right) + A_1 \sin\left(\frac{\pi}{2}(2-\tau)\right)\right] = A_2 \sin\left(\frac{\pi}{2}\tau\right)$$
$$Y_1(\tau) = \frac{1}{2}\left[A_1 \sin\left(\frac{\pi}{2}\tau\right) - A_1 \sin\left(\frac{\pi}{2}(2-\tau)\right)\right] = 0$$

for the sine and

$$X_2(\tau) = \frac{1}{2}\left[A_2 \cos\left(\frac{\pi}{2}\tau\right) + A_2 \cos\left(\frac{\pi}{2}(2-\tau)\right)\right] = 0$$
$$Y_2(\tau) = \frac{1}{2}\left[A_2 \cos\left(\frac{\pi}{2}\tau\right) - A_2 \cos\left(\frac{\pi}{2}(2-\tau)\right)\right] = A_2 \cos\left(\frac{\pi}{2}\tau\right)$$

for the cosine, respectively, where $\tau = \tau(\theta)$. Therefore

$$x_1(t) = A_1 \sin\left(\frac{\pi}{2}\tau\big(4ft\big)\right)$$
$$x_2(t) = A_2 \cos\left(\frac{\pi}{2}\tau\big(4ft\big)\right)e\big(4ft\big)$$

This example brings an interesting feature of Eq. 20 to surface. In more detail, the sine function is represented using only the $X(\tau)$ part, whereas the cosine function is represented using only the $Y(\tau)e$ part. Since the sine (cosine) function is odd (even), we may thus adopt the mapping

$$\text{odd function} \longrightarrow \quad Y = 0$$
$$\text{even function} \longrightarrow \quad X = 0$$

Hence, we may think of Eq. 20 as the periodic analog of the decomposition of a general function into even and odd parts, where $X(\tau)$ and $Y(\tau)e$ are the odd and even components over a quarter of a period.

Another important result reads as follows:

A function $f(x)$, of a periodic process $x(t)$ with period normalized to $T = 4$ can be expressed in the form of a "hyperbolic complex number"

$$f(x) = f(X + Ye) = R(\tau) + I(\tau)e \tag{22}$$

where

$$R(\tau) = \frac{1}{2}\big[f(X+Y) + f(X-Y)\big] \tag{23a}$$
$$I(\tau) = \frac{1}{2}\big[f(X+Y) - f(X-Y)\big] \tag{23b}$$

with $\tau = \tau(t)$, $e = e(t) = \dot{\tau}(t)$ and $e^2 = 1$.

The first derivative of $x(t)$ is

$$\dot{x}(t) = X'(\tau)\dot{\tau} + Y'(\tau)\dot{\tau}e + Y(\tau)\dot{e} = X'(\tau)e + Y'(\tau)e^2 + Y(\tau)\dot{e}$$

Since $\dot{\tau} = e$ and $e^2 = 1$, the latter takes the form

$$\dot{x}(t) = Y'(\tau) + X'(\tau)e + Y(\tau)\dot{e} \tag{24}$$

where the right-hand side is written in terms of the "basis" $\{1, e, \dot{e}\}$[2]. When the original function $x(t)$ is continuous at the singular points $\tau = \pm 1$ (see Fig. 2), the term $Y(\tau)\dot{e}$ can be discarded. Since the function $\dot{e}(t)$ is non-zero exactly at these singularity points, as dictated by Eq. 10, the elimination of this term is satisfied by the condition

$$Y(\tau)|_{\tau=\pm 1} = 0 \tag{25}$$

Example: we have demonstrated that

$$x_1(t) = A_1 \sin\left(\frac{\pi}{2}\tau\big(4ft\big)\right) \equiv X_1(\tau)$$
$$x_2(t) = A_2 \cos\left(\frac{\pi}{2}\tau\big(4ft\big)\right) e\big(4ft\big) \equiv Y_2(\tau)e$$

Then,

$$\dot{x}_1(t) = X_1'(\tau)\dot{\tau}(4ft) = 4fX_1'(\tau)e = 4f\frac{\pi}{2}A_1 \cos\left(\frac{\pi}{2}\tau\big(4ft\big)\right) e(4ft)$$
$$\dot{x}_2(t) = Y_2'(\tau)\dot{\tau}e + Y_2(\tau)\dot{e} = 4fY_2'(\tau) = -4f\frac{\pi}{2}A_2 \sin\left(\frac{\pi}{2}\tau\big(4ft\big)\right)$$

Observe that in the case of $x_2(t)$, it holds that

$$Y_2(\tau)|_{\tau=\pm 1} = \cos(\pi/2) = 0$$

confirming the condition of Eq. 25.

Similarly, the second derivative of $x(t)$ is

$$\ddot{x}(t) = X''(\tau) + Y''(\tau)e \tag{26}$$

provided that both

$$Y(\tau)|_{\tau=\pm 1} = 0 \quad \text{and} \quad X'(\tau)|_{\tau=\pm 1} = 0 \tag{27}$$

hold.

2 Particle between rigid barriers

2.1 Equation of motion

Formally, the motion of a particle between two rigid barriers, as sketched in Fig. 1, can be described by the Newton equation

$$m\ddot{x}(t) = 0, \qquad |x| \leq D \tag{28}$$

[2]Indeed, in hyperbolic complex algebra, the elements 1, e and $\dot{e}$ are linearly independent, just as the elements 1 and j are in the set of complex numbers (complex algebra)

for some $x(0) = x_0$ and $\dot{x}(0) = V$. For a perfectly elastic collision, the constraint condition can be replaced by a series of periodic external impulses of equal amplitude, acting on the particle whenever $|x| = D$. Hence, Eq. 28 can be transformed into

$$m\ddot{x}(t) = -F(t) = -2p \sum_{\ell=-\infty}^{\infty} \delta(t/a + 1 - 4\ell + \phi) - \delta(t/a - 1 - 4\ell + \phi) \equiv -p\dot{e}(t/a + \phi) \tag{29}$$

We observe that in contrast to Eq. 28, which is discontinuous, Eq. 29 pertains to a typical ordinary differential equation with a discontinuous excitation. In attempting to find a solution, we will apply the Fourier series expansion and the NSTT.

2.2 Solution via Fourier series

Combining Eqs. 4 and 29, a Fourier series representation of $F(t)$ reads (see App. A.2)

$$F(t) = p \sum_{\ell=-\infty}^{\infty} j \sin\left(\pi \frac{\ell}{2}\right) e^{j\pi \frac{\ell}{2}(t/a+\phi)} \tag{30}$$

Assume that the solution can be represented as a Fourier series expansion, i.e.,

$$x(t) = \sum_{\ell=-\infty}^{\infty} X_\ell e^{j\pi \frac{\ell}{2}(t/a+\phi)} \tag{31}$$

Differentiating, plugging to Eq. 29 and going over the algebra yields

$$\sum_{\ell=-\infty}^{\infty} -m\left(\frac{\pi\ell}{2a}\right)^2 X_\ell e^{j\pi \frac{\ell}{2}(\omega t+\phi)} = -p \sum_{\ell=-\infty}^{\infty} j \sin\left(\pi \frac{\ell}{2}\right) e^{j\pi \frac{\ell}{2}(\omega t+\phi)} \tag{32}$$

from which

$$X_\ell = \frac{p}{m}\left(\frac{2a}{\pi}\right)^2 \frac{\sin\left(\pi \frac{\ell}{2}\right)}{\ell^2} j \tag{33}$$

Using Eqs. 5–7, the solution can be expressed in terms of ordinary Fourier series. Since X_ℓ is complex, all A_ℓ's are zero (no cosine terms),

$$B_\ell = -2\mathrm{Im}\{X_\ell\} = -\frac{8a^2 p}{m\pi^2} \frac{\sin\left(\pi \frac{\ell}{2}\right)}{\ell^2}$$

and Eq. 31 is truncated into a sine-only representation as

$$x(t) = -\frac{8a^2 p}{m\pi^2} \sum_{\ell=1}^{\infty} \frac{\sin\left(\pi \frac{\ell}{2}\right)}{\ell^2} \sin\left(\pi \frac{\ell}{2}(t/a + \phi)\right) \tag{34}$$

Observe that

$$\sin\left(\pi \frac{\ell}{2}\right) = \begin{cases} 0 & , \ell = 2, 4, 6, \ldots \\ (-1)^{\frac{\ell-1}{2}} & , \ell = 1, 3, 5, \ldots \end{cases}$$

Therefore

$$x(t) = -\frac{a^2 p}{m} \left\{ \frac{8}{\pi^2} \sum_{\ell=1,3,5,\ldots}^{\infty} \frac{(-1)^{\frac{\ell-1}{2}}}{\ell^2} \sin\left(\pi \frac{\ell}{2}(t/a + \phi)\right) \right\} \tag{35}$$

The quantity in the brackets corresponds to the Fourier series of a time-scaled, time-shifted triangular wave of unit amplitude. Thus,

$$x(t) = -\frac{a^2 p}{m} \left\{ \frac{2}{\pi} \arcsin\left(\sin\left(\frac{\pi}{2}\frac{t}{a} + \phi\right)\right) \right\} \equiv -\frac{a^2 p}{m} \tau(t/a + \phi) \tag{36}$$

The impact intensity p can be found from the condition

$$x = \pm D \longleftrightarrow \tau = \pm 1$$

from which $p = -mD/a^2$. Substituting to Eq. 36 provides the final solution

$$x(t) = \frac{2D}{\pi} \arcsin\left(\sin\left(\frac{\pi}{2}\frac{t}{a} + \phi\right)\right) = D\tau(t/a + \phi) \tag{37}$$

which is identical to Eq. 18a. The quantities a and ϕ are recovered from the initial conditions. The latter has been already established in Eq. 19. Regarding the former, note that

$$\dot{x}(t) = D\dot{\tau}(t/a + \phi) \equiv V \tag{38}$$

and from Eq. 18a

$$D\dot{\tau}(t/a + \phi) = \frac{2D}{a\pi} \frac{\pi/2}{\sqrt{1 - \sin^2\left(\frac{\pi}{2}\frac{t}{a} + \phi\right)}} \cos\left(\frac{\pi}{2}\frac{t}{a} + \phi\right) = V \tag{39}$$

leading to

$$a = \frac{D}{V} \tag{40}$$

i.e., as defined in Sec. 1.2.1.

2.3 Solution via NSTT

Returning to Eq. 29, notice that the function $\dot{e}$ is odd. Hence, we can assume that this property is inherited to $x(t)$, allowing thus to truncate Eq. 20 as

$$x(t) = X(\tau) \tag{41}$$

with $X(-\tau) = -X(\tau)$, where $\tau = \tau(t/a + \phi)$. The second derivative of $x(t)$ reads

$$\ddot{x}(t) = \frac{1}{a^2}X''(\tau) + \frac{1}{a^2}X'(\tau)\dot{e} \tag{42}$$

Plugging to Eq. 29 and re-arranging yields

$$\frac{m}{a^2}X''(\tau) + \left(\frac{m}{a^2}X'(\tau) - p\right)\dot{e} = 0 \tag{43}$$

Based on the linear independence of 1 and $\dot{e}$, Eq. 43 is equivalent to the following, one-point boundary value problem

$$\frac{m}{a^2}X''(\tau) = 0 \quad \text{with} \quad X'(\tau)|_{\tau=1} = \frac{a^2 p}{m} \quad \text{and} \quad X(-\tau) = -X(\tau) \tag{44}$$

the solution of which is

$$X(\tau) = C_1\tau + C_2 \tag{45}$$

The oddness of $X(\tau)$ implies $C_2 = 0$, while C_1 is found from the boundary condition

$$X'(\tau)|_{\tau=1} = C_1 = \frac{a^2 p}{m} \tag{46}$$

Therefore,

$$x(t) = X(\tau) = \frac{a^2 p}{m}\tau(t/a + \phi) \tag{47}$$

which is identical to the Fourier-based one i.e., Eq. 36. As such, the parameters p, a and ϕ are calculated as described in the previous section.

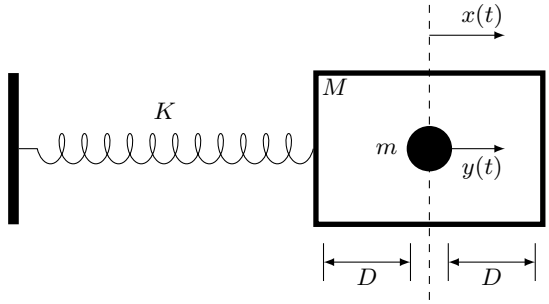


Figure 4: The impact damper.

3 The impact damper

3.1 Equations of motion

Consider now the impact damper of Fig. 4. The differential equations that describe the motion of the inertial elements are

$$M\ddot{x}(t) + Kx(t) = 0 \tag{48a}$$

$$m\ddot{y}(t) = 0, \qquad |x - y| \leq D \tag{48b}$$

Following Sec. 2, the constraint condition can be replaced by a series of external impulses, acting on both the container and the particle whenever $|x - y| = D$. Hence, the above set of equations can be transformed into

$$M\ddot{x}(t) + Kx(t) = \quad F(t) \tag{49a}$$

$$m\ddot{y}(t) = -F(t) \tag{49b}$$

where $F(t)$ is given by Eq. 30. Defining

$$r(t) = \frac{Mx(t) + my(t)}{M + m}, \qquad u(t) = x(t) - y(t) \tag{50}$$

allows transforming the set of Eqs. 49 to (refer to Sec. A.3)

$$\ddot{r}(t) + \omega_0^2\Big(r(t) + \mu u(t)\Big) = 0 \tag{51a}$$

$$\ddot{u}(t) + \omega_1^2\Big(r(t) + \mu u(t)\Big) = \frac{1}{\mu^*}F(t) \tag{51b}$$

for

$$\omega_0^2 = \frac{K}{M + m}, \qquad \omega_1^2 = \frac{K}{M}, \qquad \mu = \frac{m}{M + m}, \qquad \mu^* = \frac{Mm}{M + m} = M\mu \tag{52}$$

The above vibro-impact problem will be solved for arbitrary initial conditions of the states using (i) Fourier analysis and (ii) the NSTT.

3.2 Solution via Fourier series

By adopting Eq. 30 as a Fourier series representation for $F(t)$, the solution can be expressed as

$$r(t) = \sum_{\ell=-\infty}^{\infty} R_\ell e^{j\pi\frac{\ell}{2}(t/a+\phi)} \tag{53a}$$

$$u(t) = \sum_{\ell=-\infty}^{\infty} U_\ell e^{j\pi\frac{\ell}{2}(t/a+\phi)} \tag{53b}$$

Differentiating, plugging to Eqs. 52 and going over the algebra yields (see App. A.4)

$$\left[-\left(\frac{\pi\ell}{2a}\right)^2+\omega_0^2\right]R_\ell+\omega_0^2\mu U_\ell=0 \tag{54a}$$

$$\left[-\left(\frac{\pi\ell}{2a}\right)^2+\omega_1^2\mu\right]U_\ell+\omega_1^2R_\ell=\frac{pj}{\mu^*}\sin\left(\pi\frac{\ell}{2}\right) \tag{54b}$$

This is a typical $[2\times 2]$ linear system of equations that admits the solution (see App. A.4)

$$R_\ell=-\frac{\omega_0^2\mu p}{\mu^*\left(\frac{\pi\ell}{2a}\right)^2\left[\left(\frac{\pi\ell}{2a}\right)^2-(\omega_0^2+\omega_1^2\mu)\right]}\sin\left(\pi\frac{\ell}{2}\right)j \tag{55a}$$

$$U_\ell=\frac{\left[\left(\frac{\pi\ell}{2a}\right)^2-\omega_0^2\right]p}{\mu^*\left(\frac{\pi\ell}{2a}\right)^2\left[\left(\frac{\pi\ell}{2a}\right)^2-(\omega_0^2+\omega_1^2\mu)\right]}\sin\left(\pi\frac{\ell}{2}\right)j \tag{55b}$$

holding for all $\ell\neq 0$. Observe that U_ℓ is related to R_ℓ by (see App. A.4)

$$U_\ell=-\frac{4a^2p}{\pi^2\mu^*}\frac{\sin\left(\pi\frac{\ell}{2}\right)}{\ell^2}j+\left(\frac{\omega_1}{\omega_0}\right)^2R_\ell \tag{56}$$

and that $\omega_0^2+\omega_1^2\mu=\omega_1^2$. Hence, in terms of ordinary Fourier series, the solution admits a sine-only representation as

$$r(t)=2\frac{\omega_0^2\mu p}{\mu^*}\sum_{\ell=1}^{\infty}\frac{\sin\left(\pi\frac{\ell}{2}\right)}{\left(\frac{\pi\ell}{2a}\right)^2\left[\left(\frac{\pi\ell}{2a}\right)^2-\omega_1^2\right]}\sin\left(\pi\frac{\ell}{2}(t/a+\phi)\right)$$

$$u(t)=8\frac{a^2p}{\pi^2\mu^*}\sum_{\ell=1}^{\infty}\frac{\sin\left(\pi\frac{\ell}{2}\right)}{\ell^2}\sin\left(\pi\frac{\ell}{2}(t/a+\phi)\right)+\left(\frac{\omega_1}{\omega_0}\right)^2r(t)$$

The latter can be further manipulated as

$$r(t)=-32\frac{\omega_0^2\mu a^4p}{\mu^*\pi^4}\sum_{\ell=1}^{\infty}\frac{\sin\left(\pi\frac{\ell}{2}\right)}{\ell^2\left[\left(\frac{2a\omega_1}{\pi}\right)^2-\ell^2\right]}\sin\left(\pi\frac{\ell}{2}(t/a+\phi)\right) \tag{57a}$$

$$u(t)=\frac{a^2p}{\mu^*}\left\{\frac{8}{\pi^2}\sum_{\ell=1}^{\infty}\frac{\sin\left(\pi\frac{\ell}{2}\right)}{\ell^2}\sin\left(\pi\frac{\ell}{2}(t/a+\phi)\right)\right\}+\left(\frac{\omega_1}{\omega_0}\right)^2r(t) \tag{57b}$$

or

$$r(t)=-32\frac{\omega_0^2\mu a^4p}{\mu^*\pi^4}\sum_{\ell=1,3,5,\ldots}^{\infty}\frac{(-1)^{\frac{\ell-1}{2}}}{\ell^2\left[\left(\frac{2a\omega_1}{\pi}\right)^2-\ell^2\right]}\sin\left(\pi\frac{\ell}{2}(t/a+\phi)\right) \tag{58a}$$

$$u(t)=\frac{a^2p}{\mu^*}\left\{\frac{8}{\pi^2}\sum_{\ell=1,3,5,\ldots}^{\infty}\frac{(-1)^{\frac{\ell-1}{2}}}{\ell^2}\sin\left(\pi\frac{\ell}{2}(t/a+\phi)\right)\right\}+\left(\frac{\omega_1}{\omega_0}\right)^2r(t) \tag{58b}$$

whereas Eq. 58b can be compactly written as

$$u(t)=\frac{a^2p}{\mu^*}\tau(t/a+\phi)+\left(\frac{\omega_1}{\omega_0}\right)^2r(t) \tag{59}$$

Validation: Eq. 59 can be written as (see App. A.5)

$$my(t) = -a^2 p\tau(t/a + \phi) \tag{60}$$

Differentiating twice produces Eq. 49b.

We proceed by calculating the parameters p, ϕ and a. The former is found from the condition

$$u(t) = \pm D \longleftrightarrow \tau = \pm 1$$

Substituting to Eq. 59

$$D = -\frac{a^2 p}{\mu^*} + \left(\frac{\omega_1}{\omega_0}\right)^2 r(t_D) \tag{61}$$

where t_D denotes the time instants of the (positive) impacts (in the sense of Fig. 2). These are determined from Eq. 29 and the respective series expansion of the impact force. Clearly

$$t_D = \frac{t}{a} + \phi = 4n - 1, \qquad n = 0, \pm 1, \pm 2, \ldots$$

Hence, from Eq. 58a

$$r(t_D) = 32\frac{\omega_0^2 \mu a^4 p}{\mu^* \pi^4} \sum_{\ell=1,3,5,\ldots}^{\infty} \frac{(-1)^{\frac{\ell-1}{2}}}{\ell^2\left[\left(\frac{2a\omega_1}{\pi}\right)^2 - \ell^2\right]} \sin\left(\ell(4n-1)\frac{\pi}{2}\right) \tag{62}$$

It is easy to confirm that for the domain of ℓ and n, the far right sinusoid equals

$$\sin\left(\ell(4n-1)\frac{\pi}{2}\right) = (-1)^{\frac{\ell+1}{2}}$$

Plugging to Eq. 62 yields

$$r(t_D) = -32\frac{\omega_0^2 \mu a^4 p}{\mu^* \pi^4} \sum_{\ell=1,3,5,\ldots}^{\infty} \frac{1}{\ell^2\left[\left(\frac{2a\omega_1}{\pi}\right)^2 - \ell^2\right]} = 32\frac{\omega_0^2 \mu a^4 p}{\mu^* \pi^4} \sum_{\ell=1,3,5,\ldots}^{\infty} \frac{1}{\ell^2\left[\ell^2 - \left(\frac{2a\omega_1}{\pi}\right)^2\right]}$$

The infinite sum converges to (see App. A.6)

$$\sum_{\ell=1,3,5,\ldots}^{\infty} \frac{1}{\ell^2\left[\ell^2 - \left(\frac{2a\omega_1}{\pi}\right)^2\right]} = -\frac{\pi^4}{32a^2\omega_1^2} \tag{63}$$

with which $r(t_D)$ reduces to

$$r(t_D) = -\left(\frac{\omega_0}{\omega_1}\right)^2 \frac{\mu}{\mu^*} a^2 p$$

Substituting to Eq. 61 implies

$$D = -\frac{1}{\mu^*} a^2 p - \frac{\mu}{\mu^*} a^2 p$$

from which

$$p = -\frac{\mu^* D}{(1+\mu)a^2} \tag{64}$$

where the minus indicates that the sign of the force on the right hand side of Eqs. 49 is interchanged.

With the resulted value of p Eq. 60 becomes

$$y(t) = \frac{\mu^* D}{1+\mu}\tau(t/a + \phi)$$

For $t = 0$,

$$\tau(\phi) = \frac{1+\mu}{\mu^* D} y(0) \longrightarrow \phi = \frac{(1+\mu)\pi}{2\mu^* D} y(0) \tag{65}$$

Accordingly, differentiating Eq. 60 reads

$$\dot{y}(t) = \frac{\mu^* D}{(1+\mu)a}$$

and setting $t = 0$ gives

$$a = \frac{\mu^*}{(1+\mu)} \frac{D}{V} \tag{66}$$

where $V = \dot{y}(0)$.

3.3 Solution via NSTT

We assume a solution of the form

$$r(t) = R(\tau) \tag{67a}$$

$$u(t) = U(\tau) \tag{67b}$$

for $\tau = \tau(t/a + \phi)$. Substituting to Eqs. 51 and applying the properties of NSTT implies the boundary value problem (see App. A.7)

$$\frac{1}{a^2} R''(\tau) + \omega_0^2 R(\tau) + \omega_0^2 \mu U(\tau) = 0 \tag{68a}$$

$$\frac{1}{a^2} U''(\tau) + \omega_1^2 R(\tau) + \omega_1^2 \mu U(\tau) = 0 \tag{68b}$$

with

$$R'(\tau)|_{\tau=\pm 1} = 0 \tag{69a}$$

$$U'(\tau)|_{\tau=\pm 1} = -\frac{a^2}{\mu^*} p \tag{69b}$$

The form of the solution is (see App. A.8)

$$R(\tau) = \frac{a^2 p}{M+m} \left\{ \tau - \frac{\sin(a\omega_1 \tau)}{a\omega_1 \cos(a\omega_1)} \right\} \tag{70a}$$

$$U(\tau) = -\frac{a^2 p}{m} \left\{ \tau + \frac{m}{M} \frac{\sin(a\omega_1 \tau)}{a\omega_1 \cos(a\omega_1)} \right\} \tag{70b}$$

while it holds that (see App. A.9)

$$U(\tau) = -\frac{a^2 p}{\mu^*} \tau + \frac{1}{1-\mu} R(\tau) \tag{71}$$

Then, from Eq. 67

$$r(t) = \frac{a^2 p}{M+m} \left\{ \tau(t/a+\phi) - \frac{\sin(a\omega_1 \tau(t/a+\phi))}{a\omega_1 \cos(a\omega_1)} \right\} \tag{72a}$$

$$u(t) = -\frac{a^2 p}{m} \left\{ \tau(t/a+\phi) + \frac{m}{M} \frac{\sin(a\omega_1 \tau(t/a+\phi))}{a\omega_1 \cos(a\omega_1)} \right\} \tag{72b}$$

and

$$u(t) = -\frac{a^2 p}{\mu^*} \tau(t/a+\phi) + \frac{1}{1-\mu} r(t) \tag{73}$$

The latter is identical to Eq. 59 and also produces Eq. 60. The impact intensity p is calculated from the condition

$$u(t) = \pm D \longleftrightarrow \tau = \pm 1$$

and from Eq. 72b

$$D = -\frac{a^2 p}{m}\left\{1 + \frac{m}{M}\frac{\tan(a\omega_1)}{a\omega_1}\right\}$$

or

$$p = -\frac{mMa\omega_1}{a^2(Ma\omega_1 + m\tan(a\omega_1))}D \tag{74}$$

where, again, the minus indicates that the sign of the force on the right hand side of Eqs. 49 is interchanged.

Remark: for $\tan(a\omega_1) \approx a\omega_1$, p can be approximated as

$$p \approx -\frac{mMa\omega_1}{a^2(Ma\omega_1 + ma\omega_1)}D = -\frac{\mu^*}{a^2}D \tag{75}$$

Substituting Eq. 74 to Eq. 60 implies

$$y(t) = -\frac{Ma\omega_1 D}{Ma\omega_1 + m\tan(a\omega_1)}\tau(t/a + \phi) \tag{76}$$

and for $t = 0$

$$\tau(\phi) = -\frac{Ma\omega_1 + m\tan(a\omega_1)}{Ma\omega_1 D}y(0) \rightarrow \phi = -\frac{\pi}{2}\left(1 + \frac{m}{M}\frac{\tan(a\omega_1)}{a\omega_1}\right)\frac{y(0)}{D} \tag{77}$$

Differentiating Eq. 76 gives

$$\dot{y}(t) = -\frac{M\omega_1 D}{Ma\omega_1 + m\tan(a\omega_1)}$$

Applying the approximation $\tan(a\omega_1) \approx a\omega_1$ reduces the latter to

$$\dot{y}(t) = -\frac{\mu^* D}{a}$$

and a is recovered for $t = 0$ as

$$a = -\mu^*\frac{D}{V} \tag{78}$$

where $\dot{y}(0) = V$.

3.4 A numerical simulation

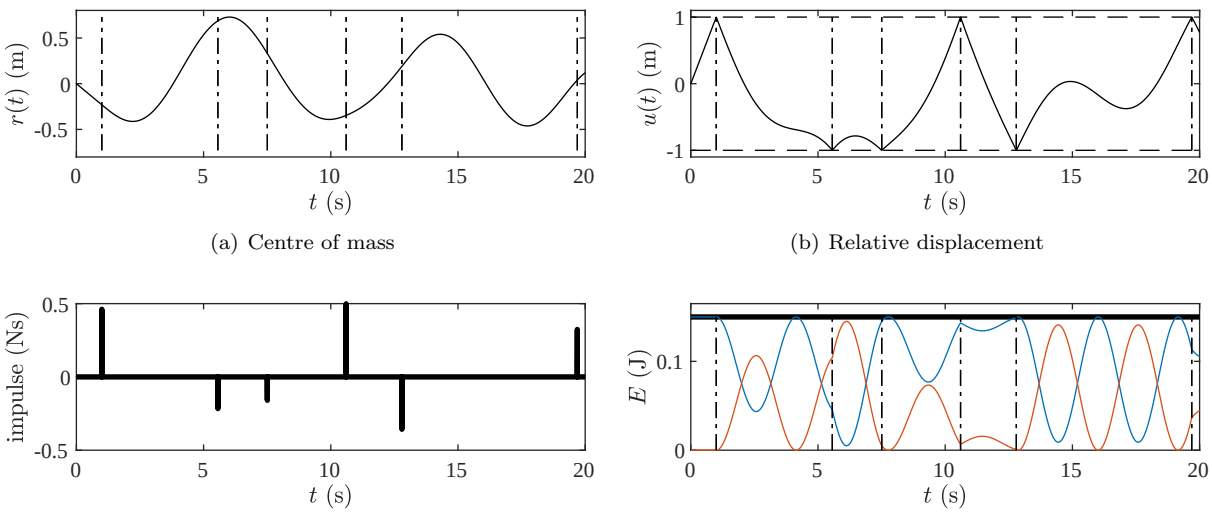
We conduct a 20 s event-based numerical simulation, by integrating the original equations of motion (Eqs. 48) for $M = 1$, $m = 0.3$, $K = 1$ and $D = 1$. We consider low, medium and high input energies, via the following initial conditions

$$\begin{aligned}
\text{low:}\quad & x(0) = 0 \quad y(0) = 0 \quad \dot{x}(0) = 0 \quad \dot{y}(0) = -1 \text{ m/s}\\
\text{medium:}\quad & x(0) = 0 \quad y(0) = 0 \quad \dot{x}(0) = 0 \quad \dot{y}(0) = -2 \text{ m/s}\\
\text{high:}\quad & x(0) = 0 \quad y(0) = 0 \quad \dot{x}(0) = 0 \quad \dot{y}(0) = -10 \text{ m/s}
\end{aligned}$$

respectively. The impulse is computed as $F(t)t_{\text{avg}} = m[\dot{y}^+(t) - \dot{y}^-(t)]$, whereas the total energy of the system is calculated at every integration step on the space of original coordinates as

$$E = \frac{1}{2}m\dot{y}^2(0) = \frac{1}{2}M\dot{x}^2(t) + \frac{1}{2}m\dot{y}^2(t) + \frac{1}{2}Kx^2(t)$$

The simulation results are expanded over Figs. 5–7. Focusing first on low energy, the most critical remark is the violation of the symmetric impacts assumption, indicating that the adopted equation for $F(t)$ is not valid. This is attributed to the fact that, in contrast to the particle case, the kinetic energy is not maintained, but transformed into potential due to the presence of the spring. This results in impacts at lower energy levels (as, indicatively, the one before the 6th second). As the input energy increases, the situation becomes better; still, though, both the periodicity and the impact intensity are not maintained constant, again controlled by the amount of energy transfer among the elastic and the inertial elements.



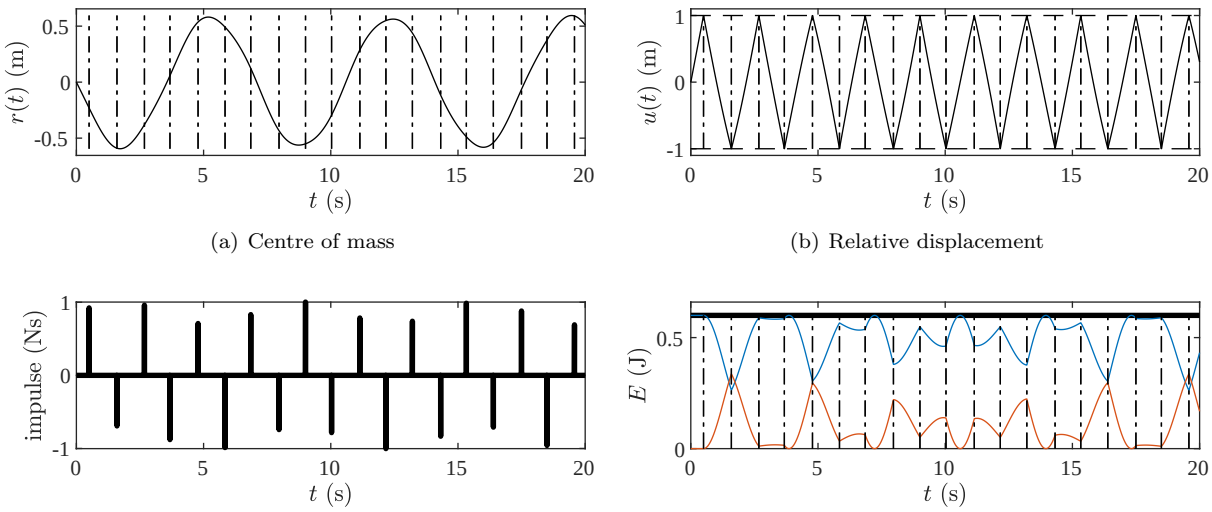
(a) Centre of mass

(b) Relative displacement

(c) Impulse

(d) Total (black), total kinetic (blue) and potential (red) energies, calculated on the space of original coordinates.

Figure 5: Impact damper: low energy simulation. The vertical dashed lines indicate the impact events.



(a) Centre of mass

(b) Relative displacement

(c) Impulse

(d) Total (black), total kinetic (blue) and potential (red) energies, calculated on the space of original coordinates.

Figure 6: Impact damper: medium energy simulation. The vertical dashed lines indicate the impact events.

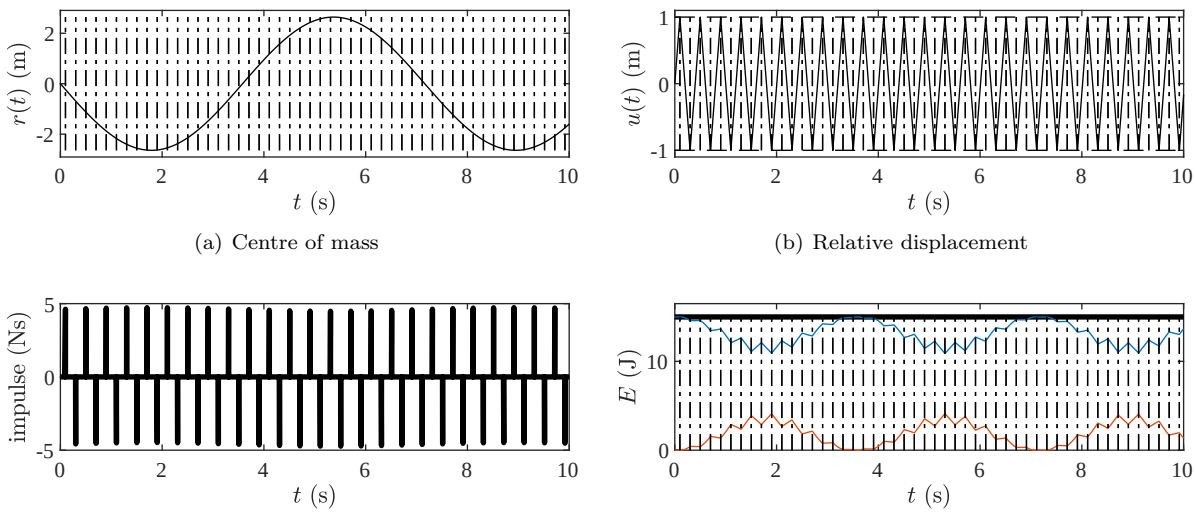
4 The meta-impactor

4.1 Equations of motion

We now turn our attention to the infinite lattice of Fig. 8, which contains an impact damper as unit cell and henceforth referred to as the *meta-impactor*. The differential equations that describe the motion of the n-th unit cell are

$$M\ddot{x}_n(t) + K\Big(2x_n(t) - x_{n-1}(t) - x_{n+1}(t)\Big) = 0 \tag{79a}$$

$$m\ddot{y}_n(t) = 0, \qquad |x_n - y_n| \leq D \tag{79b}$$



(a) Centre of mass

(b) Relative displacement

(c) Impulse

(d) Total (black), total kinetic (blue) and potential (red) energies, calculated on the space of original coordinates.

Figure 7: Impact damper: high energy simulation. The vertical dashed lines indicate the impact events.

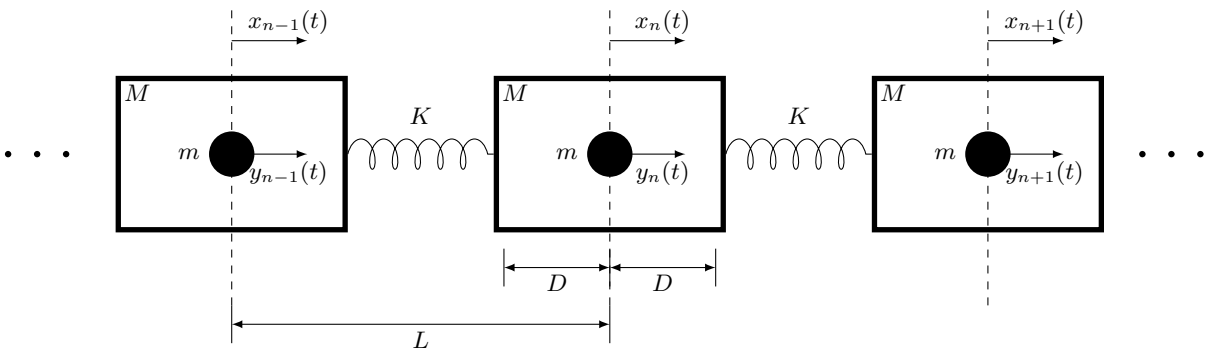


Figure 8: The meta-impactor.

Following Sec. 3.1, the constraint condition can be replaced by a series of external impulses, acting on both the container and the particle whenever $|x_n - y_n| = D$. Hence, the above set of equations can be transformed into

$$M\ddot{x}_n(t) + K\Big(2x_n(t) - x_{n-1}(t) - x_{n+1}(t)\Big) = -F_n(t) \tag{80a}$$

$$m\ddot{y}_n(t) = \quad F_n(t) \tag{80b}$$

Defining

$$r_n(t) = \frac{Mx_n(t) + my_n(t)}{M+m}, \qquad u_n(t) = x_n(t) - y_n(t) \tag{81}$$

allows transforming the set of Eq. 80 to (refer to Sec. A.10)

$$\ddot{r}_n(t) + \omega_0^2\Big\{[2r_n(t) - r_{n-1}(t) - r_{n+1}(t)] + \mu[2u_n(t) - u_{n-1}(t) - u_{n+1}(t)]\Big\} = 0 \tag{82a}$$

$$\ddot{u}_n(t) + \omega_1^2\Big\{[2r_n(t) - r_{n-1}(t) - r_{n+1}(t)] + \mu[2u_n(t) - u_{n-1}(t) - u_{n+1}(t)]\Big\} = -\frac{1}{\mu^*}F_n(t) \tag{82b}$$

where ω_0, ω_1 μ and μ^* are given by Eq. 52. Before we attempt to form a solution of the problem, we make some remarks.

Remark 1: The *eigenmodes* of the original wave equation

$$w_{tt}(x,t) = c^2 w_{xx}(x,t) \tag{83}$$

are given by

$$w(x,t) = e^{j\varphi(x,t)} = e^{j(\kappa x - \omega t)} = e^{j\kappa x} e^{-j\omega t} \tag{84}$$

A critical property of the eigenmodes reads

$$w(x + \Delta x, t) = e^{j\kappa \Delta x} w(x,t) \tag{85}$$

whereas the dispersion curve (obtained after plugging Eq. 84 to Eq. 83) is

$$\omega = c|\kappa| \tag{86}$$

Remark 2: The transition from *continuous* to *discrete* space provides the monoatomic lattice (the chain of Fig. 8 without the particles). The eigenmode at the n-th unit cell results from setting $x \to nL$, i.e.,

$$x_n(t) = e^{j(\kappa n L - \omega t)} = e^{j\kappa n L} e^{-j\omega t} \tag{87}$$

The dispersion curve (obtained after plugging Eq. 87 to Eq. 80a and setting $F_n(t) = 0$) is

$$\omega = \sqrt{\frac{4K}{M}} \sin\left(\frac{\kappa L}{2}\right) \tag{88}$$

and it is periodic in κ with period $2\pi/L$.

Remark 3: the addition of a *local resonator* to the monoatomic lattice (the chain of Fig. 8, in which the particles are connected to the containers with a linear spring of stiffness k) yields the *mass-in-mass* lattice (not to be confused with the diatomic lattice). Here, the solutions are assumed of the form

$$x_n(t) = X_k e^{j\kappa n L} e^{-j\omega t} \tag{89a}$$

$$y_n(t) = Y_k e^{j\kappa n L} e^{-j\omega t} \tag{89b}$$

The corresponding dispersion curve is characterized by an optical and an acoustic branch.

Remark 4: the common feature of all the above case is that the temporal part of the response remains unaltered.

4.2 Solution via Fourier series

4.3 Solution via NSTT

4.4 A complementary approach

A Mathematical Derivations

A.1 Proof of Equation 4

By definition (assume that $a > 0$)

$$w_s(t) = w(at \pm t_0)$$

It is straightforward to demonstrate that the period of $w_s(t)$ is $T' = T/a$, with T denoting the period of $w(t)$. The coefficients of the Fourier Series are calculated over a single period T' as

$$C_\ell = \frac{1}{T'} \int_{-T'/2}^{+T'/2} w_s(t) e^{-j2\pi \frac{\ell}{T'} t} dt$$

Substituting T' and $w_s(t)$ gives

$$C_\ell = \frac{a}{T} \int_{-T/2a}^{+T/2a} w(at \pm t_0) e^{-j2\pi \frac{a\ell}{T} t} dt = \frac{a}{T} \int_{-T/2a}^{+T/2a} w\Big(a(t \pm t_0/a)\Big) e^{-j2\pi \frac{a\ell}{T} t} dt$$

Set $t' = t \pm t_0/a$, so that

$$dt = dt' \quad \text{and} \quad t = t' \mp \frac{t_0}{a}$$

and

$$C_\ell = \frac{a}{T} \int_{-\frac{T}{2a} \pm \frac{t_0}{a}}^{+\frac{T}{2a} \pm \frac{t_0}{a}} w(at') e^{-j2\pi \frac{a\ell}{T}(t' \mp t_0/a)} dt' = \frac{a}{T} e^{\pm j2\pi \frac{\ell}{T} t_0} \int_{-T/2a}^{+T/2a} w(at') e^{-j2\pi \frac{a\ell}{T} t'} dt'$$

where the integration limits have been switched back to $\pm T/2a$, since the integration involves a single period.

We proceed by setting $t = at'$. Then $dt' = dt/a$ and

$$C_\ell = \frac{e^{\pm j2\pi \frac{\ell}{T} t_0}}{T} \cancelto{1}{\int_{-T/2}^{+T/2} w(t) e^{-j2\pi \frac{\ell}{T} t} dt} = \frac{e^{\pm j2\pi \frac{\ell}{T} t_0}}{T}$$

By definition

$$w_s(t) = \sum_{\ell=-\infty}^{\infty} C_\ell e^{j2\pi \frac{\ell}{T'} t} = \sum_{\ell=-\infty}^{\infty} C_\ell e^{j2\pi \frac{a\ell}{T} t}$$

Replacing C_ℓ by the resulted expression gives Eq. 4.

A.2 Proof of Equation 30

$F(t)$ can be written as

$$F(t) = 2p \sum_{\ell=-\infty}^{\infty} \delta(t/a + t_0 - 4\ell) - 2p \sum_{\ell=-\infty}^{\infty} \delta(t/a - t_1 - 4\ell)$$

for $t_0 = 1 + \phi$ and $t_1 = 1 - \phi$. Both individual sums are in the form of Eq. 3 with $T = 4$. Hence, based on Eq. 4, a Fourier series representation is

$$\begin{aligned} F(t) &= \frac{2p}{4} \sum_{\ell=-\infty}^{\infty} e^{j2\pi \frac{\ell}{4}(t/a + t_0)} - \frac{2p}{4} \sum_{\ell=-\infty}^{\infty} e^{j2\pi \frac{\ell}{4}(t/a - t_1)} \\ &= \frac{p}{2} \sum_{\ell=-\infty}^{\infty} e^{j\pi \frac{\ell}{2}(t/a + t_0)} - \frac{p}{2} \sum_{\ell=-\infty}^{\infty} e^{j\pi \frac{\ell}{2}(t/a - t_1)} \end{aligned}$$

The latter expression can be further manipulated as

$$F(t) = \frac{p}{2}\sum_{\ell=-\infty}^{\infty} e^{j\pi\frac{\ell}{2}t/a}\left\{e^{j\pi\frac{\ell}{2}t_0} - e^{-j\pi\frac{\ell}{2}t_1}\right\} = \frac{p}{2}\sum_{\ell=-\infty}^{\infty} e^{j\pi\frac{\ell}{2}t/a}\left\{e^{j\pi\frac{\ell}{2}(1+\phi)} - e^{-j\pi\frac{\ell}{2}(1-\phi)}\right\}$$
$$= \frac{p}{2}\sum_{\ell=-\infty}^{\infty} e^{j\pi\frac{\ell}{2}(t/a+\phi)}\left\{e^{j\pi\frac{\ell}{2}} - e^{-j\pi\frac{\ell}{2}}\right\} = p\sum_{\ell=-\infty}^{\infty} j\sin\left(\pi\frac{\ell}{2}\right)e^{j\pi\frac{\ell}{2}(t/a+\phi)}$$

A.3 Derivation of the set of Equations 51

We start from Eq. 49a and the quantity x (we drop time dependence)

$$x = \frac{1}{M+m}\left\{(M+m)x + my - my\right\} = \frac{1}{M+m}\left\{Mx + my + m(x-y)\right\} = r + \mu u$$

Equation 49a is thus written as

$$M\ddot{x}(t) + K\Big(r(t) + \mu u(t)\Big) = F(t) \tag{90}$$

Adding and subtracting $m\ddot{y}(t)$ implies

$$M\ddot{x}(t) + m\ddot{y}(t) + K\Big(r(t) + \mu u(t)\Big) - m\ddot{y}(t) = F(t)$$

Since $m\ddot{y}(t) = -F(t)$, dividing by $M+m$ leads to Eq. 51a.

We proceed with Eq. 49b

$$\begin{aligned}
m\ddot{y}(t) &= -F(t) \longrightarrow \\
m\ddot{y}(t) + m\ddot{x}(t) - m\ddot{x}(t) &= -F(t) \longrightarrow \\
-\ddot{u}(t) + \ddot{x}(t) &= -\frac{1}{m}F(t) \longrightarrow \text{ (from Eq. 90)} \\
-\ddot{u}(t) - \omega_1^2\Big(r(t) + \mu u(t)\Big) + \frac{1}{M}F(t) + \frac{1}{m}F(t) &= 0 \longrightarrow \\
\ddot{u}(t) + \omega_1^2\Big(r(t) + \mu u(t)\Big) - \frac{1}{\mu^*}F(t) &= 0
\end{aligned}$$

which is exactly Eq. 51b.

A.4 Proof of Equations 54–56

From Eqs. 53, the second derivatives of $r(t)$ and $u(t)$ can be expressed as

$$\ddot{r}(t) = -\left(\frac{\pi\ell}{2a}\right)^2\sum_{\ell=-\infty}^{\infty} R_\ell e^{j\pi\frac{\ell}{2}(t/a+\phi)}$$
$$\ddot{u}(t) = -\left(\frac{\pi\ell}{2a}\right)^2\sum_{\ell=-\infty}^{\infty} U_\ell e^{j\pi\frac{\ell}{2}(t/a+\phi)}$$

Substituting these formulas along with Eqs. 30 and 53 to Eqs. 51 and factorizing implies

$$\sum_{\ell=-\infty}^{\infty}\left\{-\left(\frac{\pi\ell}{2a}\right)^2 R_\ell + \omega_0^2 R_\ell + \omega_0^2\mu U_\ell\right\}e^{j\pi\frac{\ell}{2}(t/a+\phi)} = 0$$
$$\sum_{\ell=-\infty}^{\infty}\left\{-\left(\frac{\pi\ell}{2a}\right)^2 U_\ell + \omega_1^2 R_\ell + \omega_1^2\mu U_\ell\right\}e^{j\pi\frac{\ell}{2}(t/a+\phi)} = \sum_{\ell=-\infty}^{\infty}\frac{pj}{\mu^*}\sin\left(\pi\frac{\ell}{2}\right)e^{j\pi\frac{\ell}{2}(t/a+\phi)}$$

from which

$$
\begin{aligned}
-\left(\frac{\pi\ell}{2a}\right)^2 R_\ell + \omega_0^2 R_\ell + \omega_0^2 \mu U_\ell &= 0 \\
-\left(\frac{\pi\ell}{2a}\right)^2 U_\ell + \omega_1^2 R_\ell + \omega_1^2 \mu U_\ell &= \frac{pj}{\mu^*} \sin\left(\pi\frac{\ell}{2}\right)
\end{aligned}
$$

or

$$
\begin{aligned}
\left[-\left(\frac{\pi\ell}{2a}\right)^2 + \omega_0^2\right] R_\ell + \omega_0^2 \mu U_\ell &= 0 \\
\left[-\left(\frac{\pi\ell}{2a}\right)^2 + \omega_1^2 \mu\right] U_\ell + \omega_1^2 R_\ell &= \frac{pj}{\mu^*} \sin\left(\pi\frac{\ell}{2}\right)
\end{aligned}
$$

and Eq. 54 is proved.

The linear $[2 \times 2]$ system

$$
\begin{bmatrix}
\left[-\left(\frac{\pi\ell}{2a}\right)^2 + \omega_0^2\right] & \omega_0^2 \mu \\
\omega_1^2 & \left[-\left(\frac{\pi\ell}{2a}\right)^2 + \omega_1^2 \mu\right]
\end{bmatrix}
\begin{bmatrix} R_\ell \\ U_\ell \end{bmatrix}
=
\begin{bmatrix} 0 \\ \frac{pj}{\mu^*} \sin\left(\pi\frac{\ell}{2}\right) \end{bmatrix}
$$

is of the form

$$
\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}
\begin{bmatrix} x \\ y \end{bmatrix}
=
\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}
$$

and attains the solution

$$
\begin{aligned}
x &= \frac{c_1 b_2 - b_1 c_2}{a_1 b_2 - b_1 a_2} \\
y &= \frac{a_1 c_2 - c_1 a_2}{a_1 b_2 - b_1 a_2}
\end{aligned}
$$

Since $c_1 = 0$, the above reduce to

$$
\begin{aligned}
x &= -\frac{b_1 c_2}{a_1 b_2 - b_1 a_2} \\
y &= \frac{a_1 c_2}{a_1 b_2 - b_1 a_2}
\end{aligned}
$$

Applying these formulas we get

$$
\begin{aligned}
R_\ell &= -\frac{p}{\mu^*} \frac{\omega_0^2 \mu}{\left[-\left(\frac{\pi\ell}{2a}\right)^2 + \omega_0^2\right]\left[-\left(\frac{\pi\ell}{2a}\right)^2 + \omega_1^2 \mu\right] - \omega_0^2 \omega_1^2 \mu} j \sin\left(\pi\frac{\ell}{2}\right) \\
U_\ell &= \frac{p}{\mu^*} \frac{\left[-\left(\frac{\pi\ell}{2a}\right)^2 + \omega_0^2\right]}{\left[-\left(\frac{\pi\ell}{2a}\right)^2 + \omega_0^2\right]\left[-\left(\frac{\pi\ell}{2a}\right)^2 + \omega_1^2 \mu\right] - \omega_0^2 \omega_1^2 \mu} j \sin\left(\pi\frac{\ell}{2}\right)
\end{aligned}
$$

or,

$$
\begin{aligned}
R_\ell &= -\frac{\omega_0^2 \mu p}{\mu^* \left(\frac{\pi\ell}{2a}\right)^2 \left[\left(\frac{\pi\ell}{2a}\right)^2 - (\omega_0^2 + \omega_1^2 \mu)\right]} \sin\left(\pi\frac{\ell}{2}\right) j \\
U_\ell &= \frac{\left[-\left(\frac{\pi\ell}{2a}\right)^2 + \omega_0^2\right] p}{\mu^* \left(\frac{\pi\ell}{2a}\right)^2 \left[\left(\frac{\pi\ell}{2a}\right)^2 - (\omega_0^2 + \omega_1^2 \mu)\right]} \sin\left(\pi\frac{\ell}{2}\right) j
\end{aligned}
$$

and Eq. 55 is proved.

Equation 56 is confirmed by writing U_ℓ as

$$U_\ell = \frac{\left[-\left(\frac{\pi\ell}{2a}\right)^2 + \omega_0^2\right]p}{\mu^*\left(\frac{\pi\ell}{2a}\right)^2\left[\left(\frac{\pi\ell}{2a}\right)^2 - (\omega_0^2 + \omega_1^2\mu)\right]}\sin\left(\pi\frac{\ell}{2}\right)j$$

$$= \frac{\left[-\left(\frac{\pi\ell}{2a}\right)^2 + \omega_0^2\right] + \left[\left(\frac{\pi\ell}{2a}\right)^2 - (\omega_0^2 + \omega_1^2\mu)\right] - \left[\left(\frac{\pi\ell}{2a}\right)^2 - (\omega_0^2 + \omega_1^2\mu)\right]}{\left(\frac{\pi\ell}{2a}\right)^2\left[\left(\frac{\pi\ell}{2a}\right)^2 - (\omega_0^2 + \omega_1^2\mu)\right]}\frac{p}{\mu^*}\sin\left(\pi\frac{\ell}{2}\right)j$$

$$= -\frac{1}{\left(\frac{\pi\ell}{2a}\right)^2}\frac{p}{\mu^*}\sin\left(\pi\frac{\ell}{2}\right)j + \frac{\left[-\left(\frac{\pi\ell}{2a}\right)^2 + \omega_0^2\right] + \left[\left(\frac{\pi\ell}{2a}\right)^2 - (\omega_0^2 + \omega_1^2\mu)\right]}{\left(\frac{\pi\ell}{2a}\right)^2\left[\left(\frac{\pi\ell}{2a}\right)^2 - (\omega_0^2 + \omega_1^2\mu)\right]}\frac{p}{\mu^*}\sin\left(\pi\frac{\ell}{2}\right)j$$

$$= -\frac{4a^2p}{\pi^2\mu^*}\frac{\sin\left(\pi\frac{\ell}{2}\right)}{\ell^2}j - \frac{\omega_1^2\mu p}{\mu^*\left(\frac{\pi\ell}{2a}\right)^2\left[\left(\frac{\pi\ell}{2a}\right)^2 - (\omega_0^2 + \omega_1^2\mu)\right]}\sin\left(\pi\frac{\ell}{2}\right)j$$

$$= -\frac{4a^2p}{\pi^2\mu^*}\frac{\sin\left(\pi\frac{\ell}{2}\right)}{\ell^2}j + \left(\frac{\omega_1}{\omega_0}\right)^2 R_\ell$$

A.5 Proof of Equation 60

Start from the ratio $(\omega_1/\omega_0)^2$

$$\left(\frac{\omega_1}{\omega_0}\right)^2 = \frac{\frac{K}{M}}{\frac{K}{M+m}} = \frac{M+m}{M}$$

Then

$$\left(\frac{\omega_1}{\omega_0}\right)^2 r = \frac{M+m}{M}\frac{Mx+my}{M+m} = x + \frac{m}{M}y$$

From Eq. 59

$$u - \left(\frac{\omega_1}{\omega_0}\right)^2 r = x - y - x - \frac{m}{M}y = -\frac{M+m}{M}y$$

and thus

$$\mu^*\left[u - \left(\frac{\omega_1}{\omega_0}\right)^2 r\right] = -\frac{Mm}{M+m}\frac{M+m}{M}y = -my = a^2p\tau$$

A.6 Proof of Equation 63

For convenience set

$$\ell_0 = \frac{2a\omega_1}{\pi}$$

so that

$$\frac{1}{\ell^2\left[\ell^2 - \left(\frac{2a\omega_1}{\pi}\right)^2\right]} = \frac{1}{\ell^2(\ell^2 - \ell_0^2)} = \frac{1}{\ell^2(\ell - \ell_0)(\ell + \ell_0)}$$

A partial fraction expansion of the above quantity is

$$\frac{1}{\ell^2(\ell - \ell_0)(\ell + \ell_0)} = -\frac{1}{2\ell_0^3(\ell + \ell_0)} + \frac{1}{2\ell_0^3(\ell - \ell_0)} - \frac{1}{\ell_0^2\ell^2}$$

allowing to factorize the infinite sum of Eq. 63 as

$$\frac{1}{2\ell_0^3}\left\{\sum_{\ell=1,3,5,\ldots}^{\infty}\frac{1}{\ell-\ell_0}-\sum_{\ell=1,3,5,\ldots}^{\infty}\frac{1}{\ell+\ell_0}\right\}-\frac{1}{\ell_0^2}\sum_{\ell=1,3,5,\ldots}^{\infty}\frac{1}{\ell^2}$$

The quantity in the brackets can be modified as (note: the individual sums diverge)

$$\sum_{\ell=1,3,5,\ldots}^{\infty}\frac{1}{\ell-\ell_0}-\sum_{\ell=1,3,5,\ldots}^{\infty}\frac{1}{\ell+\ell_0}=\sum_{\ell=1}^{\infty}\left\{\frac{\sin\left(\pi\frac{\ell}{2}\right)}{\ell-\ell_0}-\frac{\sin\left(\pi\frac{\ell}{2}\right)}{\ell+\ell_0}\right\}=0$$

whereas

$$\sum_{\ell=1,3,5,\ldots}^{\infty}\frac{1}{\ell^2}=\sum_{\ell=1}^{\infty}\frac{1}{(1-2\ell)^2}=\frac{\pi^2}{8}$$

Putting it altogether, we get

$$\sum_{\ell=1,3,5,\ldots}^{\infty}\frac{1}{\ell^2(\ell^2-\ell_0^2)}=-\frac{1}{\ell_0^2}\frac{\pi^2}{8}=-\frac{\pi^4}{32a^2\omega_1^2}$$

A.7 Derivation of the set of Equations 68–69

The second derivatives of $r(t)$ and $u(t)$ are

$$\begin{aligned}\ddot{r}(t)&=\frac{1}{a^2}R''(\tau)+\frac{1}{a^2}R'(\tau)\dot{e}\\ \ddot{u}(t)&=\frac{1}{a^2}U''(\tau)+\frac{1}{a^2}U'(\tau)\dot{e}\end{aligned}$$

Substituting to Eqs. 67 gives

$$\begin{aligned}&\frac{1}{a^2}R''(\tau)+\frac{1}{a^2}R'(\tau)\dot{e}+\omega_0^2R(\tau)+\omega_0^2\mu U(\tau)=0\\ &\frac{1}{a^2}U''(\tau)+\frac{1}{a^2}U'(\tau)\dot{e}+\omega_1^2R(\tau)+\omega_1^2\mu U(\tau)=-\frac{1}{\mu^*}p\dot{e}\end{aligned}$$

Re-arranging in terms of the elements 1 and $\dot{e}$ transforms the above set of equations to

$$\begin{aligned}\left[\frac{1}{a^2}R''(\tau)+\omega_0^2R(\tau)+\omega_0^2\mu U(\tau)\right]+\left[\frac{1}{a^2}R'(\tau)\right]\dot{e}=0\\ \left[\frac{1}{a^2}U''(\tau)+\omega_1^2R(\tau)+\omega_1^2\mu U(\tau)\right]+\left[\frac{1}{a^2}U'(\tau)+\frac{1}{\mu^*}p\right]\dot{e}=0\end{aligned}$$

from which Eqs. 68–69 are established.

A.8 Derivation of the set of Equations 70

Equation 68 can be written as

$$\begin{bmatrix}R''(\tau)\\U''(\tau)\end{bmatrix}+a^2\begin{bmatrix}\omega_0^2 & \omega_0^2\mu\\ \omega_1^2 & \omega_1^2\mu\end{bmatrix}\begin{bmatrix}R(\tau)\\U(\tau)\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$$

or

$$\mathbf{z}''(\tau)+\mathbf{K}\mathbf{z}(\tau)=\mathbf{0} \tag{91}$$

where the associated vector/matrix are obviously defined. The eigenvalue problem for $\mathbf{K}$ reads

$$\mathbf{K}\mathbf{v}=\lambda\mathbf{v}$$

and from $\det\{\mathbf{K}-\lambda\mathbf{I}\}=0$ the characteristic polynomial results

$$\lambda\Big(\lambda - a^2(\omega_0^2+\omega_1^2\mu)\Big)=0$$

and its roots are

$$\lambda_1 = 0 \quad \text{and} \quad \lambda_2 = a^2(\omega_0^2+\omega_1^2\mu) = (a\omega_1)^2$$

The corresponding matrix of eigenvectors (calculated from the reduced row Echelon forms of $\mathbf{K}$ and $\mathbf{K}-\lambda_2\mathbf{I}$, respectively) is

$$\mathbf{V} = [\mathbf{v}_1 \;\; \mathbf{v}_2] = \begin{bmatrix} -\mu & \left(\frac{\omega_0}{\omega_1}\right)^2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} -\mu & 1-\mu \\ 1 & 1 \end{bmatrix}$$

Applying the transformation $\mathbf{z}=\mathbf{V}\mathbf{q}$ to Eq. 91 and sequentially left-multiplying by $\mathbf{V}^{-1}$, recovers the modal space

$$\begin{bmatrix} q_1''(\tau) \\ q_2''(\tau) \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & (a\omega_1)^2 \end{bmatrix} \begin{bmatrix} q_1(\tau) \\ q_2(\tau) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

hence,

$$\begin{aligned} q_1(\tau) &= A + B\tau \\ q_2(\tau) &= C\cos(a\omega_1\tau) + D\sin(a\omega_1\tau) \end{aligned}$$

and in the space of original quantities

$$\begin{aligned} R(\tau) &= -A\mu - B\mu\tau + (1-\mu)C\cos(a\omega_1\tau) + (1-\mu)D\sin(a\omega_1\tau) \\ U(\tau) &= \quad A + B\tau + C\cos(a\omega_1\tau) + D\sin(a\omega_1\tau) \end{aligned}$$

The constants A, B, C and D are found from the boundary conditions. Indeed, the first derivative of the solution is

$$\begin{aligned} R'(\tau) &= -B\mu - (1-\mu)a\omega_1 C\sin(a\omega_1\tau) + (1-\mu)a\omega_1 D\cos(a\omega_1\tau) \\ U'(\tau) &= \quad B - a\omega_1 C\sin(a\omega_1\tau) + a\omega_1 D\cos(a\omega_1\tau) \end{aligned}$$

This renders the parameter A free. While we may proceed by including it in the solution and calculate it through the initial conditions, recall that we already have a parameter for this, i.e., ϕ. Therefore, we can set $A=0$. Applying then Eqs. 69 results in the following system of equations

$$\begin{bmatrix} -\mu & -(1-\mu)a\omega_1\sin(a\omega_1) & (1-\mu)a\omega_1\cos(a\omega_1) \\ -\mu & (1-\mu)a\omega_1\sin(a\omega_1) & (1-\mu)a\omega_1\cos(a\omega_1) \\ 1 & -a\omega_1\sin(a\omega_1) & a\omega_1\cos(a\omega_1) \\ 1 & a\omega_1\sin(a\omega_1) & a\omega_1\cos(a\omega_1) \end{bmatrix} \begin{bmatrix} B \\ C \\ D \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -\frac{a^2p}{\mu^*} \\ -\frac{a^2p}{\mu^*} \end{bmatrix}$$

from which

$$B = -(1-\mu)\frac{a^2p}{\mu^*}, \quad C = \quad 0 \quad \text{and} \quad D = -\frac{\mu}{a\omega_1\cos(a\omega_1)}\frac{a^2p}{\mu^*}$$

and the solution reads

$$\begin{aligned} R(\tau) &= -\mu\left\{ -(1-\mu)\frac{a^2p}{\mu^*}\tau + \frac{1-\mu}{a\omega_1\cos(a\omega_1)}\frac{a^2p}{\mu^*}\sin(a\omega_1\tau)\right\} \\ U(\tau) &= \qquad -(1-\mu)\frac{a^2p}{\mu^*}\tau - \frac{\mu}{a\omega_1\cos(a\omega_1)}\frac{a^2p}{\mu^*}\sin(a\omega_1\tau) \end{aligned}$$

This can be further factorized as

$$R(\tau) = \mu(1-\mu)\frac{a^2p}{\mu^*}\left\{\tau - \frac{\sin(a\omega_1\tau)}{a\omega_1\cos(a\omega_1)}\right\} \tag{92a}$$

$$U(\tau) = -(1-\mu)\frac{a^2p}{\mu^*}\left\{\tau + \frac{\mu}{1-\mu}\frac{\sin(a\omega_1\tau)}{a\omega_1\cos(a\omega_1)}\right\} \tag{92b}$$

or

$$\begin{aligned} R(\tau) &= \frac{a^2p}{M+m}\left\{\tau - \frac{\sin(a\omega_1\tau)}{a\omega_1\cos(a\omega_1)}\right\} \\ U(\tau) &= -\frac{a^2p}{m}\left\{\tau + \frac{m}{M}\frac{\sin(a\omega_1\tau)}{a\omega_1\cos(a\omega_1)}\right\} \end{aligned}$$

A.9 Proof of Equation 71

From the set of Eqs. 92 we have

$$\begin{aligned} \frac{\sin(a\omega_1\tau)}{a\omega_1\cos(a\omega_1)} &= \tau - \frac{\mu^*}{\mu(1-\mu)a^2p}R(\tau) \\ \frac{\sin(a\omega_1\tau)}{a\omega_1\cos(a\omega_1)} &= -\frac{1-\mu}{\mu}\tau - \frac{\mu^*}{\mu a^2p}U(\tau) \end{aligned}$$

Thus

$$-\frac{1-\mu}{\mu}\tau - \frac{\mu^*}{\mu a^2p}U(\tau) = \tau - \frac{\mu^*}{\mu(1-\mu)a^2p}R(\tau)$$

Rearranging gives

$$\begin{aligned} \frac{\mu^*}{\mu a^2p}U(\tau) &= -\frac{1-\mu}{\mu}\tau - \tau + \frac{\mu^*}{\mu(1-\mu)a^2p}R(\tau) \\ &= -\frac{1}{\mu}\tau + \frac{\mu^*}{\mu(1-\mu)a^2p}R(\tau) \end{aligned}$$

finally reaching at

$$U(\tau) = -\frac{a^2p}{\mu^*}\tau + \frac{1}{1-\mu}R(\tau)$$

A.10 Derivation of the set of Equations 82

We follow the same procedure as in Sec. A.3. We start from Eq. 80a and the quantity $2x_n - x_{n-1} - x_{n+1}$ (we drop time dependence)

$$\begin{aligned} 2x_n - x_{n-1} - x_{n+1} &= \frac{1}{M+m}\Big\{2(M+m)x_n - (M+m)x_{n-1} - (M+m)x_{n+1}\Big\} \\ &= \frac{1}{M+m}\Big\{2(M+m)x_n + 2my_n - 2my_n \\ &\qquad - (M+m)x_{n-1} + my_{n-1} - my_{n-1} \\ &\qquad - (M+m)x_{n+1} + my_{n+1} - my_{n+1}\Big\} \\ &= \frac{1}{M+m}\Big\{2(Mx_n + my_n) + 2m(x_n - y_n) \\ &\qquad - (Mx_{n-1} + my_{n-1}) - m(x_{n-1} - y_{n-1}) \\ &\qquad - (Mx_{n+1} + my_{n+1}) - m(x_{n+1} - y_{n+1})\Big\} \\ &= 2r_n - r_{n-1} - r_{n+1} + \mu(2u_n - u_{n-1} - u_{n+1}) \end{aligned}$$

Equation 80a is thus written as

$$M\ddot{x}_n(t) + K\Big([2r_n - r_{n-1} - r_{n+1}] + \mu[2w_n - u_{n-1} - u_{n+1}]\Big) + F_n(t) = 0 \tag{93}$$

Substituting the force from Eq. 80b implies

$$M\ddot{x}_n(t) + K\Big([2r_n - r_{n-1} - r_{n+1}] + \mu[2u_n - u_{n-1} - u_{n+1}]\Big) + m\ddot{y}_n(t) = 0$$

and dividing by $M + m$ leads to Eq. 82a.

We proceed with Eq. 80b

$$\begin{aligned}
m\ddot{y}_n(t) - F_n(t) = 0 &\longrightarrow \\
m\ddot{y}_n(t) + m\ddot{x}_n(t) - m\ddot{x}_n(t) - F_n(t) = 0 &\longrightarrow \text{ (from Eq. 92)} \\
-m\ddot{u}_n(t) - \frac{m}{M}\Big\{K\Big([2r_n - r_{n-1} - r_{n+1}] + \mu[2w_n - u_{n-1} - u_{n+1}]\Big) + F_n(t)\Big)\Big\} - F_n(t) = 0 &\longrightarrow \\
-\ddot{u}_n(t) - \omega_1^2\Big([2r_n - r_{n-1} - r_{n+1}] + \mu[2w_n - u_{n-1} - u_{n+1}]\Big) - \left(\frac{1}{M} + \frac{1}{m}\right)F_n(t) = 0 &\longrightarrow \\
\ddot{u}_n(t) + \omega_1^2\Big\{[2r_n(t) - r_{n-1}(t) - r_{n+1}(t)] + \mu[2u_n(t) - u_{n-1}(t) - u_{n+1}(t)]\Big\} + \frac{1}{\mu^*}F_n(t) = 0 &
\end{aligned}$$

which is exactly Eq. 82b.